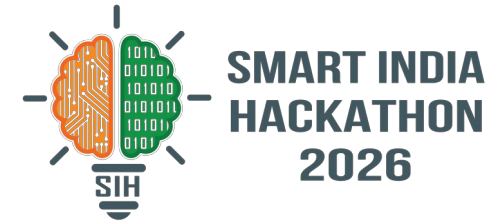


SMART INDIA HACKATHON 2026



TITLE PAGE

CryptoDrishti

An evidence-driven tool for discovering enterprise cryptography, assessing quantum exposure and planning post-quantum migration.

Problem Statement ID - SIH26164

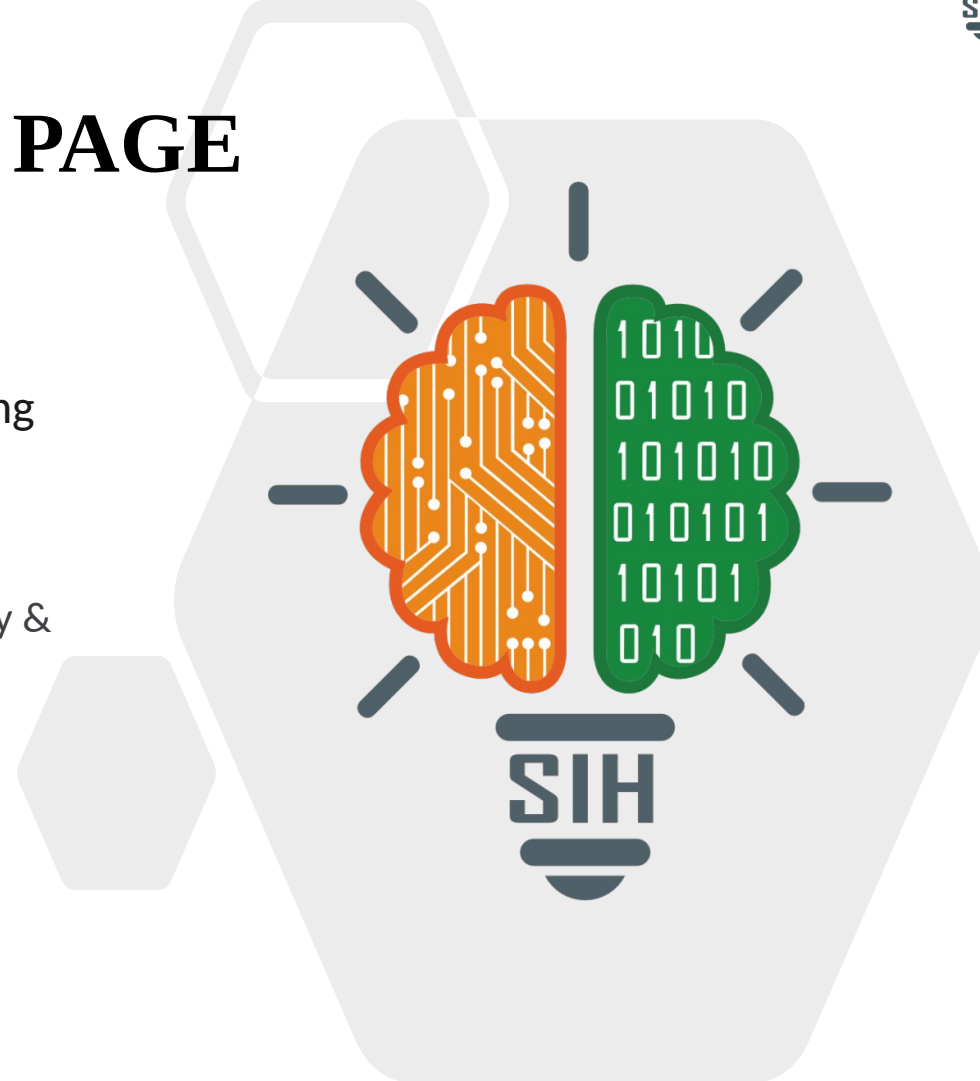
Problem Statement Title - Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

Theme - 26164

PS Category - Software

Team ID - 146876

Team Name - Zero-Day



The problem

Cryptographic assets are spread across application code, third-party libraries, configuration, certificates, compiled binaries and container images. Each records cryptography differently, and none of them is a list. Without a reliable inventory, an organisation cannot say which assets need quantum-safe migration, or in what order.

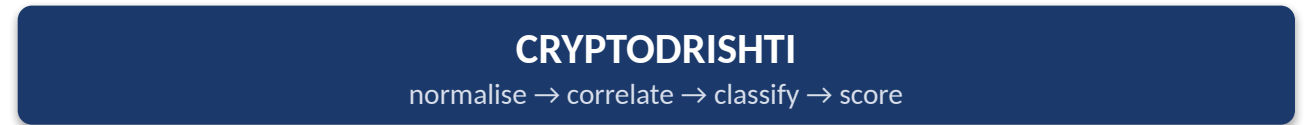
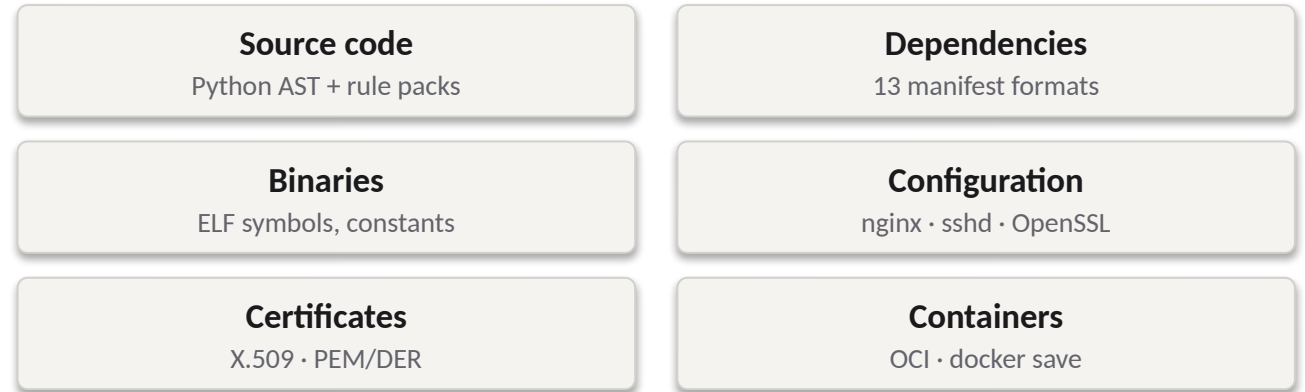
What CryptoDrishti does

CryptoDrishti discovers cryptographic evidence across those surfaces, normalises it into one inventory, assesses each asset's quantum exposure, and produces evidence-backed migration recommendations and a CycloneDX Cryptography Bill of Materials.

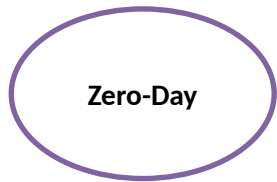
Why it is different

Finding RSA in a library is not observing RSA signing at a call site — capability, declaration, use and observation stay apart.

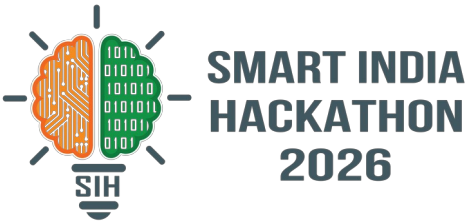
WHERE THE EVIDENCE COMES FROM



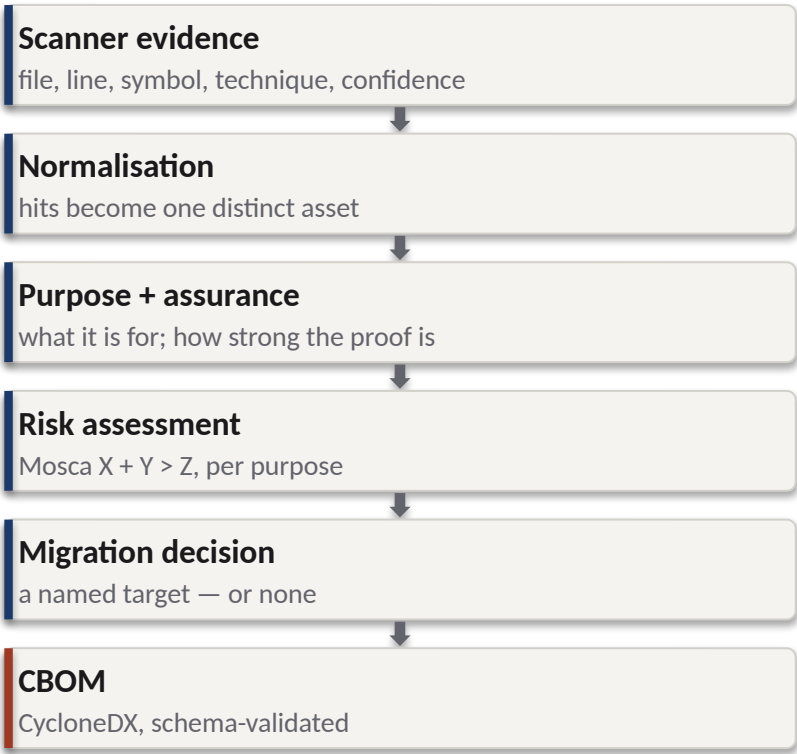
A seventh sensor completes a live TLS handshake — only against an endpoint the operator names, and only after the destination policy vets every resolved address.



TECHNICAL APPROACH

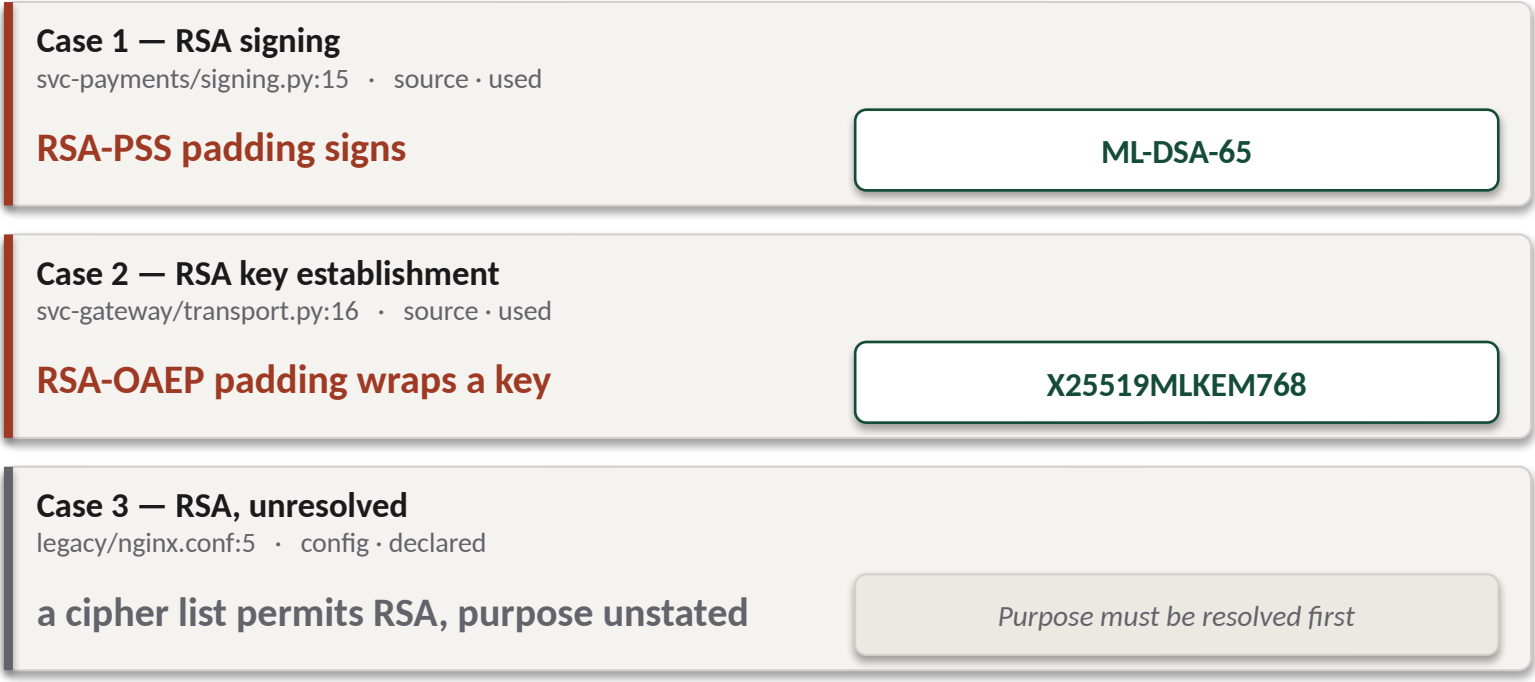


How a finding is processed



ONE ALGORITHM • THREE PURPOSE STATES • THREE DECISIONS

Every scanner can tell you RSA is present. What decides the migration is what it is being used for.



BUILT WITH

Python 3.11+ • FastAPI

SQLite (WAL)

Vanilla JS console

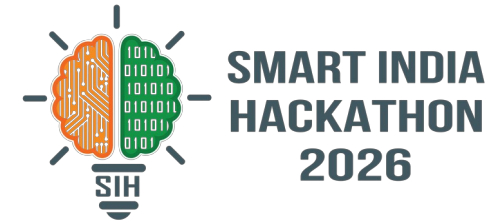
60 rules • 8 languages

ELF symbol parsing

CycloneDX 1.6 / 1.7



FEASIBILITY AND VIABILITY



Implemented

- Seven scanning surfaces, each returning evidence with a location, a technique and a confidence
- An inventory of distinct assets, carrying purpose and assurance on every one
- A quantum-risk model taking operator-supplied shelf life, criticality and constraints, saved per asset and surviving a rescan

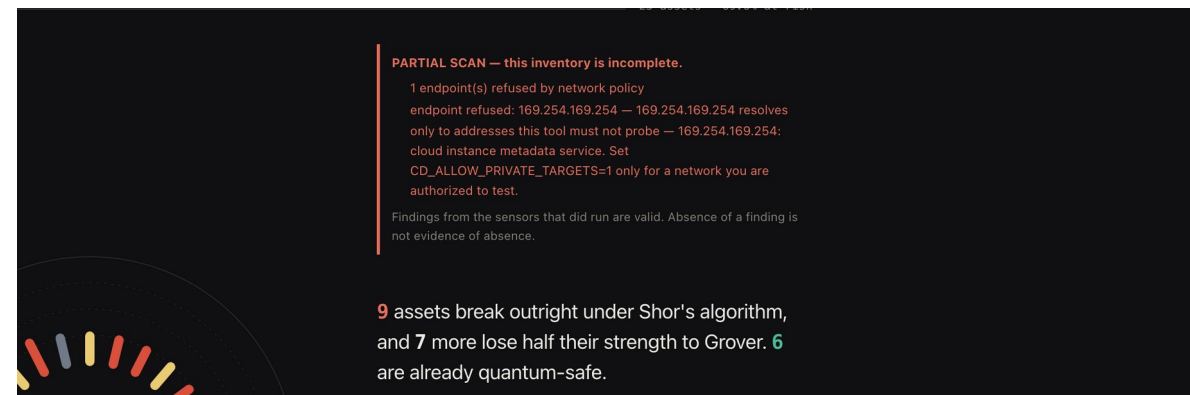
Validated

- **664 automated tests** — on Python 3.11, 3.12 and 3.13 in CI
- **Official schema validation** — every CBOM checked offline against the CycloneDX schemas at pinned commit 0bd48c8; CI fails on a violation
- **Hand-labelled benchmark** — 116 findings over six scanners, labelled from the fixtures, not from the tool's output

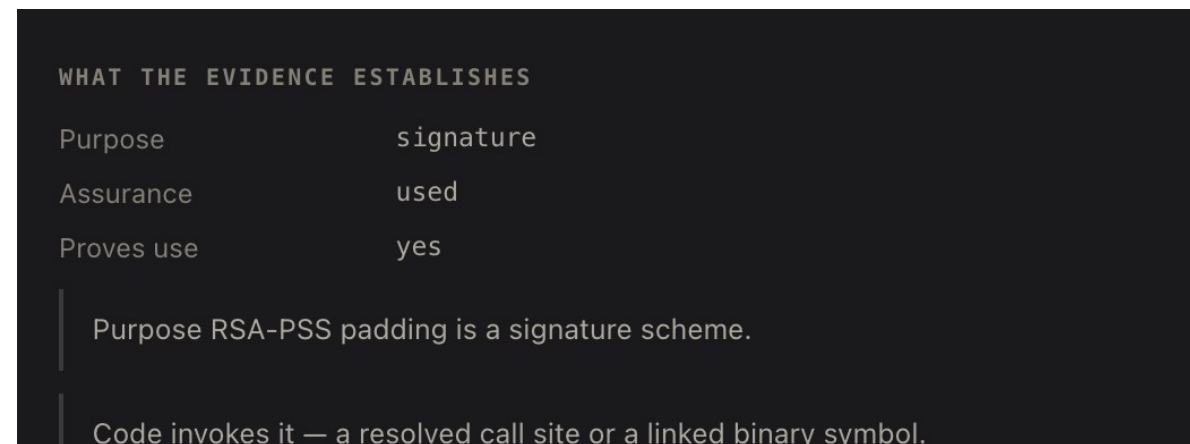
Feasible to deploy

The tool is local-first: one machine, loopback by default, no cloud service, and it validates its own output offline. Committed demo fixtures reproduce every figure in this deck with one command.

Limits — Full AST analysis is Python only; seven further languages use rule packs, and Rust and Swift are recognised but effectively uncovered. Binary parsing is ELF only. No KMS, HSM, cloud or Kubernetes integration. A working prototype, not a deployed product.



Assessment — the scan states its own completeness first: one endpoint refused, so the inventory is marked PARTIAL.



Evidence drawer — purpose, assurance and the exposure arithmetic behind one finding's score.

What a migration team does with the output



Who this is for

Intended for security engineering teams, cryptographic inventory and migration-planning teams, and infrastructure operators. These are intended users; the tool has not been deployed in production.

What changes for them

- “We think we use RSA somewhere” becomes a per-asset list with file, line, purpose, evidence strength and a named replacement.
- What an estate runs is separated from what it merely has installed, so the plan is not padded with libraries nobody calls.
- A partial scan is labelled partial, so nobody signs off an estate on an inventory that silently missed half of it.

Measured on a labelled corpus

Like-for-like, same corpus, before → after

F1 0.941 → 0.983

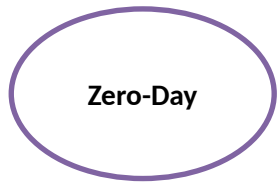
79/5/5 → 84 TP · 3 FP · 0 FN

An expanded synthetic corpus of 116 findings across six scanners scores 1.000 — a separate, non-comparable experiment. Both corpora were written by this project's developers and measure those corpora only; real-world enterprise accuracy has not been measured.

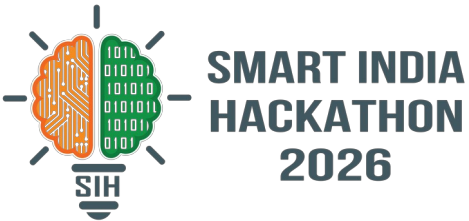
Remediation programme — grouped by the replacement each asset needs, which is how the work is staffed.

Grouped by replacement algorithm, which is how a migration is actually started and budgeted.

WORKSTREAM	ASSETS	CALL SITES	A BYTES	EFFORT	ACTION
■ ML-DSA-65	2	2	—	medium	Replace signing and verification with ML-DSA-65, and re-issue any certificates that chain to this key.
■ X25519MLKE768	2	2	—	low	Enable the hybrid group in the TLS configuration and prefer it in the group list; no application code change is required.
■ SHA-256	3	3	0	medium	Replace with SHA-256.
■ Purpose must be resolved first	3	5	—	medium	Determine what the key does, then re-run the recommendation. The quickest signals: the padding scheme (PSS signs, OAEP encrypts).



RESEARCH AND REFERENCES



Cryptography standards informing classification and migration

- FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM). NIST, 2024. doi.org/10.6028/NIST.FIPS.203
- FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA). NIST, 2024. doi.org/10.6028/NIST.FIPS.204
- FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA). NIST, 2024. doi.org/10.6028/NIST.FIPS.205
- NIST IR 8547 (initial public draft) — Transition to Post-Quantum Cryptography Standards. NIST, 2024.

Problem statement and repository

- Smart India Hackathon 2026, Problem Statement SIH26164 — Enterprise Cryptographic Discovery & Analysis Tool (ECDAT), National Technical Research Organisation (NTRO).
- Source, 664 tests, the labelled benchmark corpus and its committed results, and the vendored schemas: github.com/sgtsujith141-wq/cryptodrishti

CBOM and container formats

- CycloneDX v1.6, standardised as ECMA-424 (1st edition, June 2024) — cyclonedx.org/specification/overview
- CycloneDX v1.7 JSON Schema. Both schemas vendored in this repository at pinned commit 0bd48c8 and checksummed.
- OCI Image Format Specification — image layout, manifests and layer whiteouts. github.com/opencontainers/image-spec
- RFC 8996 — Deprecating TLS 1.0 and TLS 1.1. rfc-editor.org/rfc/rfc8996

Risk methodology

- M. Mosca, “Cybersecurity in an Era with Quantum Computers: Will We Be Ready?”, IEEE Security & Privacy, vol. 16, no. 5, 2018 — the source of the $X + Y > Z$ inequality used by the risk model.
- P. W. Shor, “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”, SIAM J. Computing, 1997.
- L. K. Grover, “A fast quantum mechanical algorithm for database search”, STOC 1996.

WHICH STANDARD GOVERNS WHICH PART OF THE PRODUCT

FIPS 203 / 204 / 205 the migration targets it names	NIST IR 8547 how it classifies what is at risk	CycloneDX 1.6 / 1.7 the CBOM it exports and validates	OCI image spec how it reads container layers	Mosca $X + Y > Z$ how it scores exposure
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Scope — CryptoDrishti discovers, classifies, assesses and recommends cryptography. It does not implement the post-quantum primitives above; it uses them as the standards that inform classification and migration targets.